

OMKAR DHAMGUNDE

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SUMMARY

Computer Science student with hands-on experience building real-time financial data systems, security applications, and scalable backend services.

Passionate about applying software engineering to financial technology – built a live trading platform processing market data across NSE and US equities with WebSocket streaming and optimized caching.

Strong foundation in DSA (1000+ LeetCode problems), agile development practices, and end-to-end project ownership from design to deployment.

EDUCATION

Vishwakarma Institute of Technology

2023 – 2027

Bachelor of Technology in Computer Science - **CGPA: 8.66**

Pune, India

TECHNICAL SKILLS

Languages: Java, Python, C, SQL, HTML, CSS

Frameworks & Tools: FastAPI, Flask, SQLAlchemy, Pandas, Numpy, Git, GitHub, MySQL

Core Concepts: DSA, OOP, DBMS, Operating Systems, API Design, Agile Development

TECHNICAL EXPERIENCE (PROJECTS)

Trading & Portfolio Analysis Platform | Python, Flask, MySQL, WebSocket, yfinance API



- Architected real-time market data streaming using **Flask-SocketIO with gevent** greenlets, pushing live price updates every 10 seconds for **130+ NSE and US equities** via yfinance batch fetching.
- Built a custom **thread-safe LRU cache (TTL=10s)** reducing external API calls by 60%, with **atomic MySQL transactions** (commit/rollback) for trade execution and dynamic P&L computation from trade history.
- Implemented Google OAuth 2.0, virtual wallet system with transaction logging, and deployed on **Gunicorn with gevent workers** for concurrent multi-user handling across financial data endpoints.

Behavioral Bot Detection | Python, FastAPI, React, TypeScript, scikit-learn



- Built a cybersecurity system using a **Random Forest classifier** trained on mouse behavioral features (velocity, acceleration, entropy), **achieving 85–95% detection accuracy** with **<5% false positive rate**.
- Designed a weighted scoring engine combining ML predictions (70%) with browser fingerprinting (30%), tested against **Selenium, Puppeteer, and headless browser** attack vectors.
- Architected full stack with **FastAPI + Uvicorn (ASGI)** serving predictions in **<100ms**, React 18 frontend with Recharts, and **SQLAlchemy + SQLite** for session storage and model retraining.

LEADERSHIP & COMMUNITY IMPACT

Problem Solving & DSA

Ongoing

LeetCode

1000+ Problems

- Solved **1000+** DSA problems on LeetCode and attained **Knight** rank with a peak rating of 1987, showcasing advanced algorithmic thinking and competitive programming skills.

Digital Literacy Initiative - Technology Educator

2024

Community Outreach & Ownership

Pune, Maharashtra

- Designed and delivered technology awareness curriculum for **50+ elderly individuals** across multiple community centers, demonstrating end-to-end ownership and inclusive leadership.
- Taught essential digital skills including UPI payments, Google Maps, and online government services, **achieving 90%+ participant proficiency** in core digital literacy.